
Project Title

HealthHub.Ai: Unified AI-Driven Health & Fitness Hub

Theme: Artificial Intelligence

Project Overview

Problem: Today's health apps are fragmented—users juggle separate tools for fitness, diet, medical records, and doctor access. AI risk assessments often lack transparency, and paper prescriptions are easily lost or altered.

Solution: HealthHub.Ai consolidates six core modules into a single web application:

- **Explainable Risk Prediction** for multiple diseases (Diabetes, Heart Disease, Cancer, Kidney/Liver Disease, Malaria, Pneumonia) using ML/DL + SHAP.
- **AI Personal Trainer** with real-time MediaPipe pose analysis for safe, guided workouts.
- **GenAI Diet Planner** that generates customized 7-day meal plans via LangChain-orchestrated LLMs.
- **Find-a-Doctor** directory with in-app appointment requests.
- **OCR Prescription Scanner** to digitize and schedule medication from images (PaddleOCR).
- **Immutable Health Records** by hashing reports on Polygon Amoy blockchain (web3.py).

Technologies: FastAPI backend, Next.js/Tailwind or Streamlit frontend, Firebase Auth & Supabase, scikit-learn/XGBoost/CNNs, Hugging Face Transformers, LangChain, MediaPipe, PaddleOCR, web3.py.

Target Users / Customers

- **Individuals:** Health-conscious adults, fitness enthusiasts, patients managing chronic conditions.
 - **Healthcare Professionals:** Doctors reviewing patient histories, prescribing follow-up, monitoring risk.
 - **Fitness Coaches & Dietitians:** Offering virtual guidance built on AI insights.
 - **Institutions:** Clinics or wellness centers seeking an integrated digital health platform.
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Key Features / Functionalities

1. **Multi-Disease Risk Predictor** – Intake of lifestyle/genetic factors → model scores + SHAP explainability.
 2. **AI Personal Trainer** – Live webcam workout with rep-count and posture feedback via MediaPipe.
 3. **GenAI Diet Planner** – LangChain-powered LLM generates personalized, calorie-tracked meal plans.
 4. **Find-a-Doctor** – Search by specialty/location → request appointments; Doctor Panel for schedule management.
 5. **OCR Prescription Scanner** – Upload Rx images → editable text extraction → schedule generation.
 6. **Blockchain Health Log** – SHA-256 hash of reports stored on Polygon Amoy for tamper-proof records.
 7. **User Dashboard** – Unified view of risk scores, workout logs, diet compliance, upcoming appointments.
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User Needs Addressed

- **Pain 1:** Switching among multiple apps; lack of cohesive health insights.
- **Pain 2:** Distrust in AI predictions without explanations.
- **Pain 3:** Lost or illegible paper prescriptions.

Solution: One-stop platform delivering **transparent, actionable** health intelligence, secure records, and professional care access.

Purpose of the Application

- **Educate & Empower:** Provide explainable AI insights to drive preventive health behaviors.
 - **Automate Manual Processes:** Replace paper logs and fragmented tools with a unified digital workflow.
 - **Enable Remote Wellness Services:** Virtual personal training, diet planning, and tele-referrals to doctors.
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Challenges You Anticipate

1. **User Trust in AI:** Mitigate by integrating SHAP explainability and transparent dashboards.
 2. **Data Accuracy:** Use validated public datasets and robust input validation.
 3. **Real-Time Performance:** Optimize CV pipeline (downsample frames, async processing) and cache ML models.
 4. **API Rate Limits:** Build local LLM fallback (Ollama) if Hugging Face quotas are reached.
 5. **Blockchain Latency:** Provide preview of hash before on-chain write; batch transactions when needed.
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Technology Stack

- **Frontend:** Next.js + Tailwind CSS *or* Streamlit for rapid prototyping
 - **Backend:** FastAPI (Python)
 - **Auth & DB:** Firebase Auth & Supabase
 - **ML/DL:** scikit-learn, XGBoost, TensorFlow/Keras (CNNs)
 - **Explainability:** SHAP
 - **GenAI:** Hugging Face Transformers, LangChain, LangGraph
 - **CV & OCR:** MediaPipe Pose, PaddleOCR
 - **Blockchain:** web3.py, Polygon Amoy testnet
 - **CI/CD & Hosting:** GitHub Actions → Streamlit Cloud / Vercel / Render
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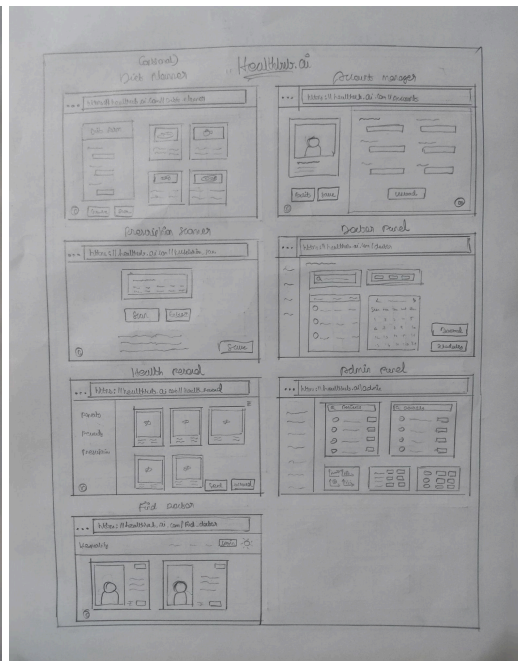
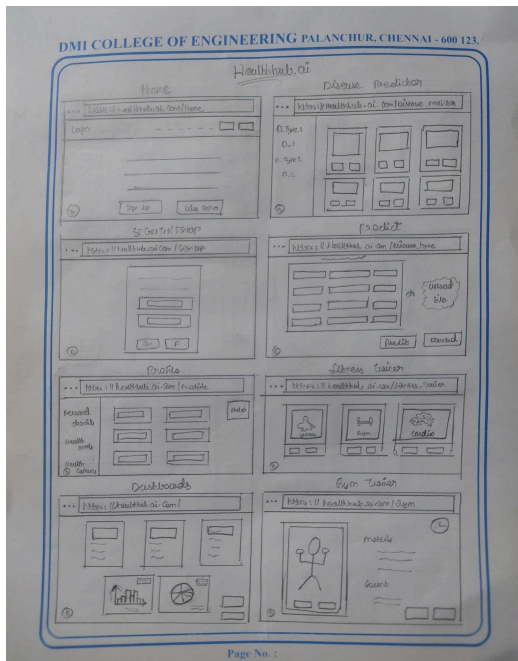
Market Research / Competitor Insights

- **MyFitnessPal / Fitbit:** Strong in diet/workout but lacks integrated medical risk analytics and blockchain security. [link](#)
- **Ada Health / Buoy Health:** Good symptom checkers but no fitness or diet planning. [link](#)

Notable Gap: Unified, explainable AI platform spanning prevention, fitness, nutrition, and secure records.

UI/UX Design Preferences

- **Theme:** “Sage Serenity” – Lighter Sage (#B2AC88), Deeper Sage (#4D5D53), Cream (#FFF3E0), Blush Pink (#F8D2F9)
- **Layout:** Clean, minimalistic cards & modals; dark/light toggle
- **Navigation:** Sidebar with clear icons for each module; responsive mobile-first design
- **Visuals:** Data-heavy charts (Plotly), live video overlays, editable forms



Content Needs

- **Educational Articles:** Blog posts explaining disease risk factors and healthy habits.
- **User Guides:** Step-by-step tutorials on using each module (risk predictor, trainer, diet planner).
- **Case Studies:** Success stories showcasing improved health outcomes.
- **API Docs:** Swagger/OpenAPI for developer reference.